

Homework Template

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Problem 1

Let $C \subset \{0, 1\}^4$ be

$$C = \{0000, 0101, 1010, 1111\}.$$

(a)

The minimum distance $d(C)$ is the minimum Hamming distance between distinct codewords.

Pairwise distances:

	0000	0101	1010	1111
0000	—	2	2	4
0101		—	4	2
1010			—	2

The smallest distance is 2. Hence

$$d(C) = 2.$$

(b)

With minimum distance d ,

$$\text{detect up to } d - 1 \text{ errors,} \quad \text{correct up to } \left\lfloor \frac{d - 1}{2} \right\rfloor.$$

Here $d = 2$, so the code can

detect at most 1 bit error, and correct 0 bit errors.

Problem 2

Let

$$C = \{(0, 0, 1), (1, 1, 1), (1, 0, 0), (0, 1, 0)\}.$$

(a) **Non-linearity**

A linear code over $\mathbb{F}_2$ must contain the zero vector. Here $000 \notin C$, so C is not linear.

(Equivalently, $(0, 0, 1) + (1, 0, 0) = (1, 0, 1) \notin C$, so C is not closed under addition.)

(b) Minimum distance and capability

Compute pairwise Hamming distances among the four codewords:

$$001, 111, 100, 010.$$

All pairwise distances are either 2 or 3; the minimum is 2. Thus

$$d(C) = 2.$$

So this code can detect at most 1 bit error and can correct 0 bit errors.

Problem 3

Binary symmetric channel with crossover probability $p < \frac{1}{2}$. Code

$$C = \{00000, 11100, 00111, 11011\},$$

received word $y = 00100$.

For a codeword x , $\Pr(y | x)$ on a BSC is

$$\Pr(y | x) = p^{d(x,y)}(1-p)^{5-d(x,y)},$$

where $d(x, y)$ is the Hamming distance.

(a) Probabilities

Distances:

$$d(00000, 00100) = 1,$$

$$d(11100, 00100) = 2,$$

$$d(00111, 00100) = 2,$$

$$d(11011, 00100) = 5.$$

So

$$\Pr(y | 00000) = p(1-p)^4,$$

$$\Pr(y | 11100) = p^2(1-p)^3,$$

$$\Pr(y | 00111) = p^2(1-p)^3,$$

$$\Pr(y | 11011) = p^5.$$

(b) Maximum likelihood estimate

Because $p < \frac{1}{2}$, the probability is largest for the smallest distance. The unique closest codeword is 00000 with distance 1.

So the ML estimate of the transmitted word is

$$\boxed{00000}.$$

Problem 4

Work over $\mathbb{F}_2$. Generator matrix

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix}.$$

Let $C = \{mG : m \in \mathbb{F}_2^2\}$.

(a) Number of codewords

$\dim C = 2$, so

$$|C| = 2^2 = 4.$$

(b) List all codewords

Messages $m = (m_1, m_2) \in \mathbb{F}_2^2$:

m	mG
$(0, 0)$	00000
$(1, 0)$	10110
$(0, 1)$	01101
$(1, 1)$	11011

So

$$C = \{00000, 10110, 01101, 11011\}.$$

(c) Minimum distance and capability

Since C is linear, $d(C)$ is the minimum Hamming weight among nonzero codewords.

Weights:

$$w(10110) = 3, \quad w(01101) = 3, \quad w(11011) = 4.$$

Thus

$$d(C) = 3.$$

This code can detect up to $d - 1 = 2$ errors and can correct

$$\left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{2}{2} \right\rfloor = 1$$

bit error.

(d) Parity-check matrix

G is in standard form $G = [I_2 \ P]$ with

$$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

A parity-check matrix is

$$H = [P^\top \ I_3] = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

Then

$$C = \{c \in \mathbb{F}_2^5 : Hc^\top = 0\}.$$

Problem 5

The parity-check matrix of a binary code C is

$$H = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

(a)

H has 5 columns, so $n = 5$. Its three rows are linearly independent over $\mathbb{F}_2$, so $\text{rank}(H) = 3$. For a binary $[n, k]$ code,

$$k = n - \text{rank}(H) = 5 - 3 = 2.$$

So the code is $[5, 2]$.

(b)

A generator matrix G must satisfy $HG^T = 0$. Let a codeword be $x = (x_1, \dots, x_5)$. From $Hx^T = 0$ we get

$$\begin{cases} x_1 + x_2 + x_3 = 0, \\ x_1 + x_4 = 0, \\ x_2 + x_5 = 0, \end{cases}$$

so

$$x_4 = x_1, \quad x_5 = x_2, \quad x_3 = x_1 + x_2.$$

Taking x_1, x_2 as free variables:

- If $(x_1, x_2) = (1, 0)$, then $x = (1, 0, 1, 1, 0)$; - If $(x_1, x_2) = (0, 1)$, then $x = (0, 1, 1, 0, 1)$.

These form a basis, so one possible generator matrix is

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

(c)

All codewords are the $\mathbb{F}_2$ -linear combinations of the rows of G :

$$\begin{aligned} (0, 0) &\rightarrow 00000, \\ (1, 0) &\rightarrow 10110, \\ (0, 1) &\rightarrow 01101, \\ (1, 1) &\rightarrow 11011. \end{aligned}$$

So

$$C = \{00000, 10110, 01101, 11011\}.$$

(d)

The code rate is

$$R = \frac{k}{n} = \frac{2}{5}.$$

Problem 6

Consider the $(7, 4)$ Hamming code over $\mathbb{F}_2$ with generator and parity-check matrices

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}.$$

Received words:

$$y_1 = 0100111, \quad y_2 = 0101010.$$

The syndrome is $s = Hy^T \pmod{2}$.

First word

For $y_1 = (0, 1, 0, 0, 1, 1, 1)$,

$$s_1 = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

The columns of H are

$$(1, 1, 0), (1, 0, 1), (0, 1, 1), (1, 1, 1), (1, 0, 0), (0, 1, 0), (0, 0, 1).$$

The syndrome $(0, 1, 0)^T$ equals column 6, so the error is in bit 6. Flip bit 6:

$$\hat{c}_1 = 0100101.$$

Since $G = [I_4 \mid P]$, the first four bits are the message:

$$m_1 = 0100.$$

Second word

For $y_2 = (0, 1, 0, 1, 0, 1, 0)$, we get

$$s_2 = Hy_2^T = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

so there is no error and y_2 is already a codeword:

$$\hat{c}_2 = 0101010, \quad m_2 = 0101.$$

Thus the corrected codewords and messages are

$$\boxed{\hat{c}_1 = 0100101, \quad m_1 = 0100; \quad \hat{c}_2 = 0101010, \quad m_2 = 0101.}$$

Problem 7

Random variables:

- X : fair coin, $\Pr(X = H) = \Pr(X = T) = \frac{1}{2}$. - Y : biased coin, $\Pr(Y = H) = \frac{3}{4}$, $\Pr(Y = T) = \frac{1}{4}$. - $Z \in \{a, b, c\}$ with $\Pr(Z = a) = \frac{1}{2}$, $\Pr(Z = b) = \frac{1}{4}$, $\Pr(Z = c) = \frac{1}{4}$.

Entropy (in bits) is $H(W) = -\sum p_i \log_2 p_i$.

(a) Entropies

$$H(X) = -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{2} \log_2 \frac{1}{2}\right) = 1.$$

$$\begin{aligned} H(Y) &= -\left(\frac{3}{4} \log_2 \frac{3}{4} + \frac{1}{4} \log_2 \frac{1}{4}\right) \\ &= \frac{3}{4} \log_2 \frac{4}{3} + \frac{1}{4} \cdot 2 \approx 0.811 \text{ bits.} \end{aligned}$$

$$\begin{aligned} H(Z) &= -\left(\frac{1}{2} \log_2 \frac{1}{2} + \frac{1}{4} \log_2 \frac{1}{4} + \frac{1}{4} \log_2 \frac{1}{4}\right) \\ &= \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 2 = 1.5 \text{ bits.} \end{aligned}$$

(b) Comparison

Numerically,

$$H(Z) = 1.5 > H(X) = 1 > H(Y) \approx 0.81.$$

So Z is the most “uncertain” (three outcomes, fairly spread), X is in the middle (fair coin), and Y is the least uncertain (heavily biased toward heads). This matches the intuitive idea that the more skewed the probabilities, the smaller the entropy.

Problem 8

Random variable X takes values in $\mathcal{A} = \{A, B, C, D, E\}$ with

$$\Pr(A) = 0.40, \quad \Pr(B) = 0.20, \quad \Pr(C) = 0.20, \quad \Pr(D) = 0.10, \quad \Pr(E) = 0.10.$$

(a) Shannon entropy

$$\begin{aligned}
H(X) &= - \sum_x p(x) \log_2 p(x) \\
&= 0.4 \log_2 \frac{1}{0.4} + 0.2 \log_2 \frac{1}{0.2} + 0.2 \log_2 \frac{1}{0.2} + 0.1 \log_2 \frac{1}{0.1} + 0.1 \log_2 \frac{1}{0.1} \\
&\approx 0.4(1.3219) + 2 \cdot 0.2(2.3219) + 2 \cdot 0.1(3.3219) \\
&\approx 2.12 \text{ bits.}
\end{aligned}$$

(b) Huffman code and average length

Probabilities: 0.40, 0.20, 0.20, 0.10, 0.10.

Huffman steps:

1. Combine $D(0.10)$ and $E(0.10) \Rightarrow DE(0.20)$. 2. Combine $B(0.20)$ and $C(0.20) \Rightarrow BC(0.40)$. 3. Combine $DE(0.20)$ and $A(0.40) \Rightarrow ADE(0.60)$. 4. Combine $ADE(0.60)$ and $BC(0.40) \Rightarrow 1.0$.

Assign 0/1 arbitrarily at each split, e.g.

Symbol	Codeword
A	01
B	10
C	11
D	000
E	001

Average code length:

$$\begin{aligned}
L &= 0.4 \cdot 2 + 0.2 \cdot 2 + 0.2 \cdot 2 + 0.1 \cdot 3 + 0.1 \cdot 3 \\
&= 1.6 + 0.6 = 2.2 \text{ bits.}
\end{aligned}$$

Check the Shannon bounds:

$$H(X) \approx 2.12 \leq L = 2.2 < H(X) + 1 \approx 3.12.$$

Problem 9

Alphabet $\{\#, A, B, \dots, Z\}$ with initial dictionary

$$\# \mapsto 0, \quad A \mapsto 1, \quad \dots, \quad Z \mapsto 26.$$

Apply Lempel–Ziv–Welch (LZW) to the string

PETERPIPERPICKEDAPECKOFFPICKLEDPEPPERS#.

(a) Output codes and new dictionary entries

Start with the initial 27 entries (codes 0–26). Scan the string, each time outputting the longest dictionary word that matches the current input, then adding that word extended by the next character as a new entry.

The output code sequence is:

16, 5, 20, 5, 18, 16, 9, 27, 31, 9, 3, 11, 5, 4, 1, 27, 37, 15, 6, 32, 37, 12, 39, 46, 11, 48, 4, 27, 16, 34,

New dictionary entries (beyond the initial symbols) in order are:

Code	Entry
27	PE
28	ET
29	TE
30	ER
31	RP
32	PI
33	IP
34	PER
35	RPI
36	IC
37	CK
38	KE
39	ED
40	DA
41	AP
42	PEC
43	CKO
44	OF
45	FP
46	PIC
47	CKL
48	LE
49	EDP
50	PICK
51	KL
52	LED
53	DP
54	PEP
55	PP
56	PERS
57	S#

Codes 0–31 fit in 5 bits. Once code 32 appears, we need 6-bit codes. The first code ≥ 32 in the output is 32, so:

- First 19 codes are < 32 : $19 \times 5 = 95$ bits. - Remaining 13 codes are ≥ 32 : $13 \times 6 = 78$ bits.

Total LZW output length:

$95 + 78 = 173 \text{ bits}$

(b) Comparison with naive encoding

The original string has 45 characters. Naively encoding each character with a fixed 5-bit code (using only the initial alphabet) would use

$$45 \times 5 = 225 \text{ bits.}$$

LZW uses 173 bits, so the savings are

$$225 - 173 = \boxed{52 \text{ bits}}.$$

Problem 10

We have 16 coins, one of which is odd (either heavier or lighter), and a two-pan balance that only reports “balance” or “do not balance.” We do *not* need to know whether the odd coin is heavier or lighter, only which coin it is.

Each weighing can be viewed as a yes/no question of the form “Is the odd coin among this set of coins?” so every weighing gives one bit of information. With k weighings we can distinguish at most 2^k possibilities. Since there are 16 coins, we must have

$$2^k \geq 16 \quad \Rightarrow \quad k \geq 4.$$

So at least 4 weighings are necessary.

Strategy using 4 weighings.

Label the coins $1, 2, \dots, 16$ and write each label in 4-bit binary:

Coin	Binary label
1	0001
2	0010
3	0011
4	0100
$\vdots$	$\vdots$
16	1111

For $i = 1, 2, 3, 4$ (from least significant bit to most), perform weighing i as the yes/no test:

Weighing i : “Is the odd coin among those whose i -th bit is 1?”

Operationally, we put all coins with i -th bit 1 on the left pan and the same number of other coins (chosen so that the total number of coins on each side is the same) on the right pan. If the scale balances, the odd coin is not in this subset; if it does not balance, then the odd coin is in this subset.

Record the 4-bit pattern of answers, using

$$\text{“no”} \rightarrow 0, \quad \text{“yes”} \rightarrow 1.$$

For example, if the outcomes are

$$(\text{yes, no, yes, yes}) = 1011_2,$$

then the odd coin is the one with label $1011_2 = 11$.

Every coin has a unique 4-bit label, and the 4 yes/no answers reproduce that label, so after 4 weighings we can identify the odd coin uniquely.

Thus 4 weighings are both necessary and sufficient.

References